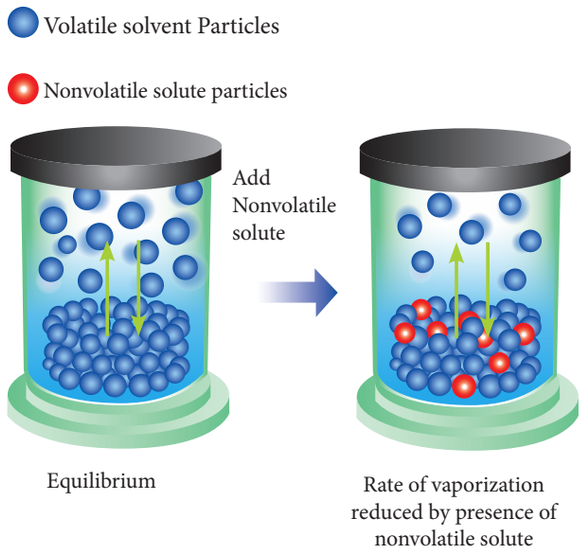


Fig 9.6 Rate of vapourization reduced by presence of nonvolatile solute.

$P_{solution} \alpha x_A$ ---------- (9.11)

Where x_A is the mole fraction of the solvent

$P_{solution} = k x_A$ ---------- (9.12)

When $x_A = 1$, $K = P^o_{solvent}$

($P^o_{solvent}$ is the partial pressure of pure solvent)

$P_{solution} = P^o_{solvent} x_A$ ----------(9.13)

$$\frac{P_{solution}}{P^o_{solvent}} = x_A \text{ ----------(9.14)}$$

$$1 - \frac{P_{solution}}{P^o_{solvent}} = 1 - x_A \text{ ----------(9.15)}$$

$$\frac{P^o_{solvent} - P_{solution}}{P^o_{solvent}} = x_B \text{ ---------- (9.16)}$$

Where x_B is the mole fraction of the solute

($\because x_A + x_B = 1, x_B = 1 - x_A$)

The above expression gives the relative lowering of vapour pressure. Based on this expression, Raoult's Law can also be stated as "the relative lowering of vapour pressure of an ideal solution containing the nonvolatile solute is equal to the mole fraction of the solute at a given temperature".

Comparison of Raoult's law and Henry's law

According to Raoult's law, for a solution containing a nonvolatile solute

$p_{solute} = p^\circ_{solute} x_{solute}$ ---------- (9.17)

According to Henry's law:

$p_{solute} = K_H x_{solute\ in\ solution}$ ---------- (9.18)

The difference between the above two expressions is the proportionality constant p°_A (Raoults Law) and K_H.(Henry's Law). Henry's law is applicable to solution containing gaseous solute in liquid solvent, while the Raoult's Law is applicable to nonvolatile solid solute in a liquid solvent. If the solute is non volatile then the Henry's law constant will become equal to the vapour pressure of the pure solvent (p°_A) and thus, Raoult's law becomes a special case of Henry's law. For very dilute solutions the solvent obeys Raoult's law and the solute obeys Henry's law.

9.8 Ideal and non-ideal solutions

9.8.1 Ideal Solutions:

An ideal solution is a solution in which each component i.e. the solute as well as the solvent obeys the Raoult's law over the entire range of concentration. Practically no solution is ideal over the entire range of concentration. However,

Unit 9.indd 45 23-12-2021 17:00:56